

Laboratory Work #2

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NORMALIZATION

Task 1: Violation Identification

The unnormalized table `Booking_receipt(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_numbers, ticket_price)` violates 1NF, 2NF, and 3NF.

- 1NF Violation: The `seat_numbers` column contains non-atomic values (comma-separated values like "12A, 12B" for multiple passengers on the same booking).
- 2NF Violation: Non-key attributes like `passenger_full_name` depend only on part of a composite primary key once normalized to 1NF.
- 3NF Violation: Transitive dependencies exist between non-key attributes. For example, `departure_city` depends on `departure_airport_name`, and `airline_name` depends on `flight_number`, rather than depending directly on `booking_id`.

Task 2: Step-by-Step Normalization

Step 1: First Normal Form (1NF)

To achieve 1NF, we remove the multi-valued `seat_numbers` column by splitting rows so each seat gets its own record. The primary key becomes composite: `(booking_id, seat_number)`.

Table: `Booking_receipt_1NF`

Attributes: `booking_id, seat_number, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price`

Primary Key: `(booking_id, seat_number)`

Step 2: Second Normal Form (2NF)

To achieve 2NF, we eliminate partial functional dependencies where attributes depend only on `booking_id` instead of the full composite key `(booking_id, seat_number)`. We split the table into `Booking_details` and `Seat_assignments`.

Table 1: `Booking_2NF`

Attributes: `booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price`

Primary Key: `booking_id`

Table 2: `Booking_Seat_2NF`

Attributes: `booking_id, seat_number`

Primary Key: `(booking_id, seat_number)`

Foreign Key: `booking_id` REFERENCES `Booking_2NF(booking_id)`

Step 3: Third Normal Form (3NF)

To achieve 3NF, we eliminate transitive dependencies where non-key attributes depend on other non-key attributes.

- passenger_full_name depends on passenger_passport_number.
- departure_city depends on departure_airport_name.
- arrival_city depends on arrival_airport_name.
- airline_name depends on flight_number.

We decompose into 6 3NF tables:

1. Table: Passenger

Attributes: passport_number, passenger_full_name

Primary Key: passport_number

2. Table: Airport

Attributes: airport_name, city

Primary Key: airport_name

3. Table: Flight

Attributes: flight_number, departure_airport_name, arrival_airport_name, airline_name

Primary Key: flight_number

Foreign Keys:

- departure_airport_name REFERENCES Airport(airport_name)
- arrival_airport_name REFERENCES Airport(airport_name)

4. Table: Booking

Attributes: booking_id, passenger_passport_number, flight_number, ticket_price

Primary Key: booking_id

Foreign Keys:

- passenger_passport_number REFERENCES Passenger(passport_number)
- flight_number REFERENCES Flight(flight_number)

5. Table: Booking_Seat

Attributes: booking_id, seat_number

Primary Key: (booking_id, seat_number)

Foreign Key: booking_id REFERENCES Booking(booking_id)

Task 3: Functional Dependencies (FDs)

The functional dependencies used to justify the normalization steps:

1. FDs for 1NF -> 2NF:

- (booking_id, seat_number) -> passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, ticket_price (Full Dependency)
- booking_id -> flight_number, ticket_price (Partial Dependency removed from composite key)

2. FDs for 2NF -> 3NF:

- passenger_passport_number -> passenger_full_name
- flight_number -> departure_airport_name, arrival_airport_name, airline_name
- departure_airport_name -> departure_city
- arrival_airport_name -> arrival_city
- booking_id -> passenger_passport_number, flight_number, ticket_price

Task 4: 3NF Analysis of the Flights Table

In the airport database created in the DDL section:

Table: Flights

Primary Key: flight_id

Functional Dependencies:

- flight_id -> sch_departure_time, sch_arrival_time, departing_airport_id, arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time, act_arrival_time, created_at, updated_at
- departing_airport_id -> airport_name, city, country (located in Airport table)
- arriving_airport_id -> airport_name, city, country (located in Airport table)
- airline_id -> airline_name, airline_code (located in Airline table)

Conclusion: The Flights table IS in 3NF. All non-key attributes depend directly and only on the primary key (flight_id). Attributes belonging to airports and airlines are stored in their respective tables (Airport, Airline) and referenced via foreign keys (departing_airport_id, arriving_airport_id, airline_id), preventing any transitive dependencies within the Flights table itself.

Task 5: DDL & Sample Data for Normalized Booking_receipt (Task 2 Tables)

```
CREATE TABLE Receipt_Passenger (  
  passport_number VARCHAR(20) PRIMARY KEY,  
  passenger_full_name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Receipt_Airport (  
airport_name VARCHAR(50) PRIMARY KEY,  
city VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Receipt_Flight (  
flight_number VARCHAR(20) PRIMARY KEY,  
departure_airport_name VARCHAR(50) NOT NULL REFERENCES  
Receipt_Airport(airport_name),  
arrival_airport_name VARCHAR(50) NOT NULL REFERENCES  
Receipt_Airport(airport_name),  
airline_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE Receipt_Booking (  
booking_id INT PRIMARY KEY,  
passenger_passport_number VARCHAR(20) NOT NULL REFERENCES  
Receipt_Passenger(passport_number),  
flight_number VARCHAR(20) NOT NULL REFERENCES  
Receipt_Flight(flight_number),  
ticket_price DECIMAL(7,2) NOT NULL  
);
```

```
CREATE TABLE Receipt_Booking_Seat (  
booking_id INT NOT NULL REFERENCES Receipt_Booking(booking_id),  
seat_number VARCHAR(10) NOT NULL,  
PRIMARY KEY (booking_id, seat_number)  
);
```

```
INSERT INTO Receipt_Passenger VALUES  
( 'N1001', 'Maksat Rakhat'),  
( 'P2002', 'John Smith'),  
( 'A3003', 'Elena Petrova'),  
( 'B4004', 'Ali Khan'),  
( 'C5055', 'Sarah Connor');
```

```
INSERT INTO Receipt_Airport VALUES  
( 'JFK International', 'New York'),  
( 'Heathrow Airport', 'London'),  
( 'Almaty International', 'Almaty'),  
( 'Astana Airport', 'Astana');
```

```
INSERT INTO Receipt_Flight VALUES  
( 'KC871', 'Almaty International', 'Heathrow Airport', 'Air Astana'),  
( 'BA123', 'Heathrow Airport', 'JFK International', 'British Airways'),  
( 'AA505', 'JFK International', 'Almaty International', 'American Airlines'),  
( 'KC992', 'Astana Airport', 'Almaty International', 'Air Astana'),  
( 'BA404', 'Heathrow Airport', 'Astana Airport', 'British Airways');
```

```
INSERT INTO Receipt_Booking VALUES
(101, 'N1001', 'KC871', 450.00),
(102, 'P2002', 'BA123', 600.00),
(103, 'A3003', 'AA505', 550.00),
(104, 'B4004', 'KC992', 120.00),
(105, 'C5055', 'BA404', 480.00);
```

```
INSERT INTO Receipt_Booking_Seat VALUES
(101, '12A'),
(101, '12B'),
(102, '14C'),
(103, '01A'),
(104, '05D'),
(105, '22F');
```

DDL

1. Initial Table Creation

```
CREATE TABLE Airline_info ( airline_id INT PRIMARY KEY, airline_code
VARCHAR(30) NOT NULL, airline_name VARCHAR(50) NOT NULL, airline_country
VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at
TIMESTAMP NOT NULL, info VARCHAR(50) NOT NULL );
```

```
CREATE TABLE Airport ( airport_id INT PRIMARY KEY, airport_name VARCHAR(50)
NOT NULL, country VARCHAR(50) NOT NULL, state VARCHAR(50) NOT NULL, city
VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at
TIMESTAMP NOT NULL );
```

```
CREATE TABLE Passengers ( passenger_id INT PRIMARY KEY, first_name
VARCHAR(50) NOT NULL, last_name VARCHAR(50) NOT NULL, date_of_birth DATE
NOT NULL, gender VARCHAR(50) NOT NULL, country_of_citizenship VARCHAR(50)
NOT NULL, country_of_residence VARCHAR(50) NOT NULL, passport_number
VARCHAR(20) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at
TIMESTAMP NOT NULL );
```

```
CREATE TABLE Flights ( flight_id INT PRIMARY KEY, sch_departure_time TIMESTAMP
NOT NULL, sch_arrival_time TIMESTAMP NOT NULL,
departing_airport_id INT NOT NULL, arriving_airport_id INT NOT NULL, departing_gate
VARCHAR(50) NOT NULL, arriving_gate VARCHAR(50) NOT NULL, airline_id INT
NOT NULL, act_departure_time TIMESTAMP NOT NULL, act_arrival_time TIMESTAMP
NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL
);
```

```
CREATE TABLE Booking ( booking_id INT PRIMARY KEY, flight_id INT NOT NULL,
passenger_id INT NOT NULL, booking_platform VARCHAR(50) NOT NULL, created_at
TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, status VARCHAR(50)
NOT NULL, price DECIMAL(7,2) NOT NULL );
```

```
CREATE TABLE Booking_flight ( booking_flight_id INT PRIMARY KEY, booking_id INT NOT NULL, flight_id INT NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL );
```

```
CREATE TABLE Boarding_pass ( boarding_pass_id INT PRIMARY KEY, booking_id INT NOT NULL, seat VARCHAR(50) NOT NULL, boarding_time TIMESTAMP NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL );
```

```
CREATE TABLE Baggage ( baggage_id INT PRIMARY KEY, weight_in_kg DECIMAL(4,2) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, booking_id INT NOT NULL );
```

```
CREATE TABLE Baggage_check ( baggage_check_id INT PRIMARY KEY, check_result VARCHAR(50) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, booking_id INT NOT NULL, passenger_id INT NOT NULL );
```

```
CREATE TABLE Security_check ( security_check_id INT PRIMARY KEY, check_result VARCHAR(20) NOT NULL, created_at TIMESTAMP NOT NULL, updated_at TIMESTAMP NOT NULL, passenger_id INT NOT NULL );
```

2. Schema Alterations (ALTER TABLE)

```
ALTER TABLE Airline_info RENAME TO Airline;
```

```
ALTER TABLE Booking RENAME COLUMN price TO ticket_price;
```

```
ALTER TABLE Flights ALTER COLUMN departing_gate TYPE TEXT;
```

```
ALTER TABLE Airline DROP COLUMN info;
```

3. Foreign Key Constraints Setup

```
ALTER TABLE Flights ADD CONSTRAINT fk_flights_departing_airport FOREIGN KEY (departing_airport_id) REFERENCES Airport(airport_id), ADD CONSTRAINT fk_flights_arriving_airport FOREIGN KEY (arriving_airport_id) REFERENCES Airport(airport_id), ADD CONSTRAINT fk_flights_airline FOREIGN KEY (airline_id) REFERENCES Airline(airline_id);
```

```
ALTER TABLE Booking ADD CONSTRAINT fk_booking_flight FOREIGN KEY (flight_id) REFERENCES Flights(flight_id), ADD CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
```

```
ALTER TABLE Booking_flight ADD CONSTRAINT fk_bf_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id), ADD CONSTRAINT fk_bf_flight FOREIGN KEY (flight_id) REFERENCES Flights(flight_id);
```

```
ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_pass_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
```

```
ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking FOREIGN KEY (booking_id) REFERENCES Booking(booking_id);
```

```
ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggage_check_booking
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id), ADD CONSTRAINT
fk_baggage_check_passenger FOREIGN KEY (passenger_id) REFERENCES
Passengers(passenger_id);
```

```
ALTER TABLE Security_check ADD CONSTRAINT fk_security_check_passenger
FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id);
```

4. Sample Data Insertion

```
INSERT INTO airport (airport_id, airport_name, country, state, city, created_at, updated_at)
VALUES (1, 'JFK Airport', 'USA', 'NY', 'New York', NOW(), NOW()), (2, 'Heathrow Airport',
'UK', 'England', 'London', NOW(), NOW());
```

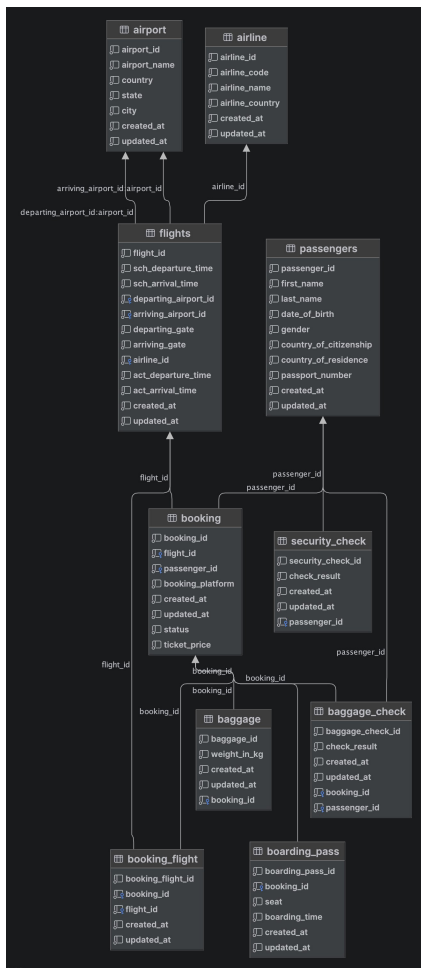
```
INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at,
updated_at) VALUES (1, 'AA', 'American Airlines', 'USA', NOW(), NOW()), (2, 'BA',
'British Airways', 'UK', NOW(), NOW());
```

```
INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender,
country_of_citizenship, country_of_residence, passport_number, created_at, updated_at)
VALUES (1, 'John', 'Smith', '1995-05-10', 'Male', 'USA', 'USA', 'A12345678', NOW(),
NOW());
```

```
INSERT INTO flights (flight_id, sch_departure_time, sch_arrival_time, departing_airport_id,
arriving_airport_id, departing_gate, arriving_gate, airline_id, act_departure_time,
act_arrival_time, created_at, updated_at) VALUES (101, '2025-10-01 10:00:00', '2025-10-01
18:00:00', 1, 2, 'Gate 12', 'Gate 5', 1, '2025-10-01 10:05:00', '2025-10-01 18:00:00', NOW(),
NOW());
```

```
INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform, created_at,
updated_at, status, ticket_price) VALUES (501, 101, 1, 'Website', NOW(), NOW(),
'Confirmed', 500.00);
```

```
INSERT INTO boarding_pass (boarding_pass_id, booking_id, seat, boarding_time,
created_at, updated_at) VALUES (701, 501, '14B', '2025-10-01 09:15:00', NOW(), NOW());
```



```
-- 1. Create Airline_info Table
CREATE TABLE AirLine_info (
    airline_id INT PRIMARY KEY,
    airline_code VARCHAR(50) NOT NULL,
    airline_name VARCHAR(50) NOT NULL,
    airline_country VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    info VARCHAR(50) NOT NULL
);

-- 2. Create Airport Table
CREATE TABLE Airport (
    airport_id INT PRIMARY KEY,
    airport_name VARCHAR(50) NOT NULL,
    country VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- 3. Create Passengers Table
CREATE TABLE Passengers (
    passenger_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    gender VARCHAR(10) NOT NULL,
    country_of_citizenship VARCHAR(50) NOT NULL,
    country_of_residence VARCHAR(50) NOT NULL,
    passport_number VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- 4. Create Flights Table
CREATE TABLE Flights (
    flight_id INT PRIMARY KEY,
    sch_departure_time TIMESTAMP NOT NULL,
    sch_arrival_time TIMESTAMP NOT NULL,
    departing_airport_id INT NOT NULL,
    arriving_airport_id INT NOT NULL,
    departing_gate VARCHAR(10) NOT NULL,
    arriving_date DATE NOT NULL,
    airline_id INT NOT NULL,
    act_departure_time TIMESTAMP NOT NULL,
    act_arrival_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- 5. Create Booking Table
CREATE TABLE Booking (
    booking_id INT PRIMARY KEY,
    flight_id INT NOT NULL,
    passenger_id INT NOT NULL,
    booking_platform VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    status VARCHAR(20) NOT NULL,
    ticket_price DECIMAL(10,2) NOT NULL
);

-- 6. Create Security Check Table
CREATE TABLE Security_check (
    security_check_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    check_result VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    passenger_id INT NOT NULL
);

-- 7. Create Baggage Table
CREATE TABLE Baggage (
    baggage_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    weight_in_kg DECIMAL(10,2) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- 8. Create Baggage Check Table
CREATE TABLE Baggage_check (
    baggage_check_id INT PRIMARY KEY,
    baggage_id INT NOT NULL,
    check_result VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL,
    booking_id INT NOT NULL,
    passenger_id INT NOT NULL
);

-- 9. Create Booking Flight Table
CREATE TABLE Booking_flight (
    booking_flight_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    flight_id INT NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- 10. Create Boarding Pass Table
CREATE TABLE Boarding_pass (
    boarding_pass_id INT PRIMARY KEY,
    booking_id INT NOT NULL,
    flight_id INT NOT NULL,
    seat VARCHAR(10) NOT NULL,
    boarding_time TIMESTAMP NOT NULL,
    created_at TIMESTAMP NOT NULL,
    updated_at TIMESTAMP NOT NULL
);

-- Add Foreign Key Constraints
ALTER TABLE Booking ADD CONSTRAINT fk_booking_flight FOREIGN KEY (flight_id) REFERENCES Flights (flight_id);
ALTER TABLE Booking ADD CONSTRAINT fk_booking_passenger FOREIGN KEY (passenger_id) REFERENCES Passengers (passenger_id);
ALTER TABLE Security_check ADD CONSTRAINT fk_security_check_booking FOREIGN KEY (booking_id) REFERENCES Booking (booking_id);
ALTER TABLE Security_check ADD CONSTRAINT fk_security_check_passenger FOREIGN KEY (passenger_id) REFERENCES Passengers (passenger_id);
ALTER TABLE Baggage ADD CONSTRAINT fk_baggage_booking FOREIGN KEY (booking_id) REFERENCES Booking (booking_id);
ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggage_check_booking FOREIGN KEY (booking_id) REFERENCES Booking (booking_id);
ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggage_check_passenger FOREIGN KEY (passenger_id) REFERENCES Passengers (passenger_id);
ALTER TABLE Baggage_check ADD CONSTRAINT fk_baggage_check_baggage FOREIGN KEY (baggage_id) REFERENCES Baggage (baggage_id);
ALTER TABLE Booking_flight ADD CONSTRAINT fk_booking_flight_booking FOREIGN KEY (booking_id) REFERENCES Booking (booking_id);
ALTER TABLE Booking_flight ADD CONSTRAINT fk_booking_flight_flight FOREIGN KEY (flight_id) REFERENCES Flights (flight_id);
ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_pass_booking FOREIGN KEY (booking_id) REFERENCES Booking (booking_id);
ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_pass_flight FOREIGN KEY (flight_id) REFERENCES Flights (flight_id);
ALTER TABLE Boarding_pass ADD CONSTRAINT fk_boarding_pass_baggage FOREIGN KEY (baggage_id) REFERENCES Baggage (baggage_id);
```

DataGrip diagrams screenshot